Theoretical Predictions and Derived Results

Spectral Tests of Dark Matter and Cosmic Structure in Quantum Information Copy Time

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This register collects the numbered predictions, derived relations and numerical benchmarks of the main article and the supplementary identities. Each entry specifies its domain and supporting calculation. Cross-references give the section, proposition, equation or table number in the compiled document named in the same row.

The entries distinguish finite QICT operator certificates, conditional physical identities, declared model parameters and calculations using observational reference inputs. The scalar-portal masses and couplings, background normalization and halo profiles retain their stated external provenance. A numerical result for a declared reference model has that model as its domain. The finite C9 certificate concerns the static chiral-source polar-unitary construction described in the main article.

The supplied finite calculations establish the spectral and response results listed below. Their physical extension uses the source, state and convergence hypotheses in Supplementary Sections S1–S4. In particular, a source-visible pole requires a nonzero residue in addition to an isolated operator eigenvalue; SI10 and SI11 give complementary linear and amplitude enclosures under verified common-limit error bounds. A QICT dark-particle mass, its relic abundance, complete primordial CMB spectra and a sample-wide galaxy fit are not established by these inputs. The support column records the tests actually performed; mathematical or numerical agreement is distinct from independent experimental confirmation.

ID	Statement or value	Domain and cross-reference	Support
Source spectra and finite operator certificates			
SP1	$\lambda_{\max}(F_{13}(0)) = 0.016084$ at R4	Main article: Table 1; Six-source C9 Fejér upper bound; finite regulator	Finite moments; exact recorded-data certificate
SP2	$\lambda_{\max}(B_{13}(0)) \simeq 0.001956273$ at R4	Main article: Table 2; Inverse-Toeplitz upper bound on the same putative atom	Finite moments; exact recorded-data certificate
SP3	R4 atomic weight ≤ 0.001956273015	Main article: Table 3; Exact rational, normalized recorded C9 moments	Finite moments; exact recorded-data certificate
SP4	Flat moment rank reconstructs all cyclic dynamics	Main article: Proposition 10; exact positivity and rank equality	Analytic proof; numerical matrix controls
SP5	$\|C_n - C_1^n\| \leq \min\{2, n(n-1)\varepsilon/2\}$	Main article: Proposition 11; unitary transfer and isometric source	Analytic proof; numerical matrix controls

ID	Statement or value	Domain and cross-reference	Support
SP6	Exponential atomic-weight enclosure	Main article: Proposition 14; independently certified angular gap	Analytic proof; numerical matrix controls
SP7	Joint regulator and moment limit	Main article: Section 2.13; uniform gap and weak convergence	Analytic proof under the stated hypotheses
SP8	Finite-arc spectral-weight bound	Main article: Eq. (2.17); stated angular resolution	Analytic proof under the stated hypotheses
SP9	Fixed-polynomial moment-error enclosure	Main article: Eq. (2.18); verified moment errors	Analytic proof under the stated hypotheses
SP10	Optimal bound decreases to the visible atom	Main article: Proposition 13; fixed positive measure	Analytic proof under the stated hypotheses
SP11	Largest feasible atomic component is B_N	Main article: Proposition 12; positive-definite Toeplitz form	Analytic proof under the stated hypotheses
SP12	$\mu_L(\{1\}) = \mu_R(\{1\}) = 0$ at R1–R4	Main article: Proposition 17; certified polar-unitary finite model	Analytic proof; outward-rounded operator certificate
SP13	R4 chord gaps $g_L \geq 1.92308 \times 10^{-7}$, $g_R \geq 3.19912 \times 10^{-4}$	Main article: Table 4; all relevant spectral pairs	Analytic proof; outward-rounded operator certificate
SP14	$F_N(0) \leq \min\{1, 4/(N^2 g^2)\}$, $\lvert G(1)\rvert \leq g^{-1}$	Main article: Eq. (3.8); normalized separated finite measure	Analytic proof; outward-rounded operator certificate
SP15	Operator errors with $\sum_j \eta_j < g$ preserve separation	Main article: Section 3.3; fixed source and spectator structure	Analytic proof; outward-rounded operator certificate
SP16	Positive finite Lehmann weights reconstruct every moment	Main article: Proposition 15; stationary state and isometric source	Analytic proof; independent moment reconstruction
Identifiability and independent operator information			
MAT1	Distinct spectral types with identical moments through any finite order N	Main article: Proposition 1; explicit normalized scalar and six-channel measures	Analytic proof or explicit construction
MAT2	Integer energy branches share one transfer unitary	Main article: Proposition 5; fixed clock	Analytic proof or explicit construction
MAT3	Identical two-point moments admit distinct four-point interactions	Main article: Proposition 6; explicit positive Hamiltonian family	Analytic proof or explicit construction
MAT4	Invisible odd-sector phases are independently variable	Main article: Proposition 3; sources supported in the even sector	Analytic proof or explicit construction
MAT5	Clock scaling gives $E \mapsto E/s$	Main article: Proposition 2; fixed dimensionless transfer	Analytic proof or explicit construction

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MAT6	$\|\rho_{\mathcal{G}} - \rho_{\mathrm{diag}}\|_1 \leq \varepsilon_{\mathrm{coh}}$	Main article: Eq. (3.3); retained joint eigenspace coherences	Analytic proof or explicit construction
Finite gravitational response			
GR1	$\epsilon_{0,\mathrm{src},R} = 5.889 \times 10^{-5}$	Main article: Section 4.4; Finite source convention: $\partial_J \ln\sqrt{-g} = 4$	Nine-state source evaluation
GR2	Connected $D(0) = 0.1494933611$; curvature 0.05084820378	Main article: Equation (4.8); Nine-state source derivatives, distinct contact term	Nine-state source evaluation
GR3	Positive response derivatives and alternating remainder	Main article: Proposition 20; positive Lehmann weights	Analytic proof; finite-response checks
GR4	$a_1 = 0.35360648$, $a_2 = 1.29642815$, analytic disk $\|s\| < 0.2209$	Main article: Eq. (4.11); supplied nine-state source	Nine-state source evaluation
GR5	Contact shift $B \mapsto B + bI$ shifts curvature by b	Main article: Eq. (4.13); fixed K_0 and linear source	Analytic proof; finite-response checks
GR6	Curvature error $\leq Mx + (\delta_h + \delta_a)/x$	Main article: Eq. (4.15); declared remainder and data errors	Analytic proof; finite-response checks
Gravitational matching and expansion			
DE1	$G = c^5\tau_*^2/(8\pi\hbar\gamma_R)$	Main article: Equation (4.2); Conditional QICT identity	Conditional physical identity
DE2	$\Lambda = \epsilon_0/(\gamma_R c^2 \tau_*^2)$	Main article: Equation (4.2); Conditional QICT identity	Conditional physical identity
DE3	$\Lambda_{\mathrm{obs}} = 1.09076 \times 10^{-52}\ \mathrm{m}^{-2}$	Main article: Equation (6.2); Observational-input reconstruction	Reconstruction from stated observational inputs
DE4	$u_{\Lambda,\mathrm{obs}} = 5.2525 \times 10^{-10}\ \mathrm{J/m}^3$	Main article: Equation (6.2); Observational-input reconstruction	Reconstruction from stated observational inputs
DE5	$p = -u_{\mathrm{vac}}$; $w = -1$ for $u_{\mathrm{vac}} \neq 0$	Main article: Proposition 19; Conditional metric-response theorem	Conditional physical identity
DE6	$\epsilon_0/\gamma_R^2 = (7.1528 \pm 0.1910) \times 10^{-121}$	Main article: Equation (4.5); Planck base-ΛCDM reference	Reconstruction from stated observational inputs
Particle interactions and thermal production			
DM1	$m_S = 10, 40, 300$ GeV (distinct examples)	Main article: Section 5; Independent effective-model inputs	Declared effective-model input
DM2	$\sigma_{\mathrm{SI}}(40) = 5.282 \times 10^{-46}\ \mathrm{cm}^2$	Main article: Section 5.1; Higgs portal benchmark; excluded; full local scalar abundance	Rate calculation; published LZ comparison
DM3	$\sigma_{\mathrm{SI}}(300) = 3.910 \times 10^{-45}\ \mathrm{cm}^2$	Main article: Section 5.1; Higgs portal benchmark	Specified thermal or scattering model calculation
DM4	$\langle\sigma v\rangle_{hh}^{\mathrm{contact}} = 2.345 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s}$	Main article: Section 5.2; Higgs-pair contact contribution	Specified thermal or scattering model calculation
DM5	$Y_{\mathrm{FI}} = 3.6292 \times 10^{-11}$	Main article: Equation (5.8); One daughter per parent decay; stated parent, width and thermal history	Specified thermal or scattering model calculation

ID	Statement or value	Domain and cross-reference	Support
DM6	$\Omega_{\mathrm{FI}}h^2 = 0.09951$	Main article: Section 5.3; Freeze-in, selected effective width	Specified thermal or scattering model calculation
DM7	$\lvert\lambda_{HS}\rvert \leq 6.454 \times 10^{-4}$ at 40 GeV	Main article: Equation (5.11); Inversion of published LZ bound in selected model	Rate calculation; published LZ comparison
DM8	$\Gamma_A^{\mathrm{req.}} = 1.205877 \times 10^{-20}$ GeV	Main article: Equation (5.12); Observation-normalized inverse calculation	Specified thermal or scattering model calculation
DM9	Constant-rate freeze-out $\Omega_\chi h^2 = 0.100322$	Main article: Equation (5.15); Numerical Boltzmann limiting equation	Specified thermal or scattering model calculation
DM10	$Y_\infty \propto \delta_A\sqrt{8\pi\gamma_R}/\varepsilon_A^2$	Main article: Equation (5.28); Exact scale cancellation in specified one-daughter limit	Analytic derivation; thermal numerical controls
DM11	$\Omega_S h^2 = 0.0234522$ at $m_S = 300$ GeV, $\lambda_{HS} = 0.20$	Main article: Equation (5.22); Thermal tree-level two-body portal, constant $g_* = 90$	Specified thermal or scattering model calculation
DM12	$I_0^{-1} \leq Y_\infty \leq (I_0 - E)^{-1}$	Main article: Proposition 21; positive source and $E < I_0$	Analytic derivation; thermal numerical controls
DM13	Decay yield with variable thermal degrees of freedom	Main article: Eq. (5.32); entropy conservation and equilibrated parent	Analytic derivation; thermal numerical controls
DM14	$\partial \ln Y/\partial \ln T_{\mathrm{RH}} = 7.78357$ at $x_{\mathrm{RH}} = 10$	Main article: Eq. (5.33); constant-degree decay model	Analytic derivation; thermal numerical controls
DM15	Odd-branch stability under a strict final-state threshold	Main article: Proposition 4; exact parity and energy conservation	Analytic stability proof
Matter growth and structure			
LSS1	$D(0.5) = 0.76894$	Main article: Section 6.2.1; Smooth-Λ baseline	Reference-model numerical calculation
LSS2	$f\sigma_8(0.5) = 0.47449$	Main article: Section 6.2.1; Baseline, observed σ_8 input	Reference-model numerical calculation
LSS3	$\mathrm{d}n/\mathrm{d}\ln M(10^{14}M_\odot) = 2.704 \times 10^{-5}$ Mpc^{-3}	Main article: Section 6.3.1; Approximate Press–Schechter	Reference-model numerical calculation
LSS4	Spherical collapse threshold $a = 0.4614218$	Main article: Section 6.3.2; Chosen overdensity and mass	Reference-model numerical calculation
LSS5	$C_{50}^{\kappa\kappa} = 1.701 \times 10^{-8}$; $C_{200}^{\kappa\kappa} = 2.544 \times 10^{-9}$	Main article: Equation (7.12); Single-source Limber/linear GR baseline	Reference-model numerical calculation
Galaxy populations and gas dynamics			
GAL1	$M_\star(10^{12}M_\odot) = 7.816 \times 10^9 M_\odot$	Main article: Section 6.3.3; Chosen efficiency $\varepsilon_\star$	Reference-model numerical calculation

ID	Statement or value	Domain and cross-reference	Support
GAL2	$\xi_g(10\,\mathrm{Mpc}) = 1.462405$ for $b_g = 1.5$	Main article: Section 6.3.3; Smooth-spectrum benchmark	Reference-model numerical calculation
GAL3	$M_g(3\,\mathrm{Gyr}) = 8.8841 \times 10^9 M_\odot$	Main article: Equation (6.19); Selected one-zone gas-regulator inputs	Reference-model numerical calculation
GAL4	$M_*(3\,\mathrm{Gyr}) = 5.4619 \times 10^9 M_\odot$	Main article: Equation (6.19); Same gas-regulator solution	Reference-model numerical calculation
GAL5	Positive gas solution for time-dependent inflow	Main article: Eq. (6.20); nonnegative rates and stated return fraction	Exact reservoir solution and balance check
GAL6	$M_w(3\,\mathrm{Gyr}) = 9.10314 \times 10^9 M_\odot$	Main article: Eq. (6.21); exact baryon balance in the selected reservoir	Exact reservoir solution and balance check
Particle–mesh evolution			
PM1	Final density RMS 10.4690; peak 1484.30	Main article: Section 6.4; Coherent-wave periodic 32^3 PM experiment	200/300-step comparison at the stated mesh
Lensing and scalar metric response			
LEN1	SIS $\theta_E = 0.8884$ arcsec	Main article: Section 7.2; Chosen geometry and dispersion	Reference-model numerical calculation
LEN2	SIS $\lvert\Delta t\rvert = 42.93$ d	Main article: Section 7.2; Chosen source offset	Reference-model numerical calculation
LEN3	NFW $\kappa(0.1\,\mathrm{Mpc}) = 0.1197$	Main article: Section 7.3; Chosen cluster mass/concentration	Reference-model numerical calculation
LEN4	NFW $\gamma_t(0.1\,\mathrm{Mpc}) = 0.0708$	Main article: Section 7.3; Chosen cluster mass/concentration	Reference-model numerical calculation
LEN5	Cluster $\theta_t = 5.35$ arcsec	Main article: Section 7.3; Separate idealized massive cluster	Reference-model numerical calculation
LEN6	Angular convergence ratio at $\ell = 200$: 1.2709505	Main article: Section 7.5; Selected scalar response and matched initial density	Reference-model numerical calculation
LEN7	$E_G(0.5) = 0.41400$ (GR), 0.40097 (selected scalar response)	Main article: Section 7.5; Declared background and growth conditions	Reference-model numerical calculation
LEN8	Growth and E_G invert to μ and η	Main article: Eq. (7.17); quasistatic conserved matter and fixed background	Analytic inversion; integrated-growth check
CMB geometry and transfer relations			
CMB1	$r_s(z_*) = 144.294$ Mpc	Main article: Section 8.1; Reference recombination input	Reference-background numerical calculation
CMB2	$100\theta_* = 1.04075$	Main article: Section 8.1; Reference, observational normalization	Reference-background numerical calculation

ID	Statement or value	Domain and cross-reference	Support
CMB3	$\ell_A = 301.86$	Main article: Section 8.1; Acoustic spacing of the stated background	Reference-background numerical calculation
CMB4	Saha $z_{x_e=1/2} = 1369$	Main article: Section 8.2; Equilibrium-only ionization	Reference-background numerical calculation
CMB5	$C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}$	Main article: Equations (8.9), (8.13); Analytic observables; primordial and transfer inputs specified separately	Transfer-functional identity or analytic proof
CMB6	$\lvert C_\ell^{TE}\rvert^2 \le C_\ell^{TT} C_\ell^{EE}$	Main article: Proposition 22; positive primordial measure	Transfer-functional identity or analytic proof
CMB7	Finite spectra admit a primordial equivalence class	Main article: Eq. (8.15); fixed transfers and positive reference spectrum	Transfer-functional identity or analytic proof
Supplementary physical identities			
SI1	$\lVert P_{\mathcal{R}} - P\rVert \le r\eta_{\mathcal{R}}$; isolated rank-one eigenvalue persists	Supplement: Proposition S1; common physical resolvents; source pole requires positive residue	Contour proof; matrix-sequence controls
SI2	$\lvert E_{\chi,\mathcal{R}} - E_\chi\rvert \le 2r^2\eta_{\mathcal{R}}$; $\lVert Z_{\mathcal{R}} - Z\rVert \le r\eta_{\mathcal{R}} + 2\varepsilon_{\mathcal{R}}$	Supplement: Equations (S2.3)–(S2.4); positive limiting source weight	Analytic bounds; independent contour integration
SI3	Connected cumulants determine $\Gamma^{(3)}$ and $\Gamma^{(4)}$	Supplement: Proposition S2; locally invertible source-to-field map	Exact cumulants; independent inverse-map derivatives
SI4	Fixed dynamics admit distinct initial abundances	Supplement: Proposition S3; specified admissible initial densities	Convex density-operator construction
SI5	$h^2 + k\ell_*^2/a^2 = (\widehat{u} + \epsilon_0)/(3\gamma_R)$	Supplement: Equation (S4.1); matched Einstein branch and cosmological state	Algebraic substitution of gravitational matching
SI6	$v_c^2 = r\Psi'$; deflection from $\Phi + \Psi$	Supplement: Equation (S4.3); common weak-field metric and stated symmetries	Timelike and null-geodesic relations
SI7	On-shell amplitude from the connected four-point function and pole residues	Supplement: Equation (S3.5); stable scalar branch, Lorentzian continuation and asymptotic states	LSZ reduction with the specified field normalization
SI8	$\mathrm{d}\sigma = \overline{\lvert\mathcal{M}\rvert^2}\mathrm{d}\Phi_f/(4FS_f)$; $F = \sqrt{(p_1\cdot p_2)^2 - m_1^2 m_2^2}$	Supplement: Equation (S3.6); specified incoming particles and final-state symmetry factor	Relativistic flux and phase-space identity
SI9	Primordial curvature covariance determines the power-spectrum input to CMB transfers	Supplement: Equation (S4.4); homogeneous isotropic initial state and specified transfer evolution	Spectral covariance definition and transfer integrals

ID	Statement or value	Domain and cross-reference	Support
SI10	$u^\dagger Z u \geq u^\dagger Z_{\mathcal{R}} u - r\eta_{\mathcal{R}} - 2\varepsilon_{\mathcal{R}}$	Supplement: Equation (S2.5); unit source vector and verified common-limit error bounds	Analytic enclosure; visible and invisible source controls
SI11	$[\max\{0, \sqrt{z_{\mathcal{R}}} - d_{\mathcal{R}}\}]^2 \leq z$ $z \leq \min\{1, (\sqrt{z_{\mathcal{R}}} + d_{\mathcal{R}})^2\}$	Supplement: Equation (S2.6); normalized source; $z = u^\dagger Z u$, $d_{\mathcal{R}} = r\eta_{\mathcal{R}} + \varepsilon_{\mathcal{R}}$	Reverse-triangle proof; weak-overlap matrix control

Reproducibility and error conventions. The numerical values, finite matrix arrays, interval certificates and formula checks are distributed in the code and data folder. Its `README.md` gives the reproduction commands and expected outputs. The exact recorded-moment certificate, outward-rounded finite-operator enclosure and floating-point numerical comparisons retain their respective arithmetic guarantees. The conditional physical identities carry the assumptions stated in their proofs. Regulator convergence and empirical error bounds enter where explicitly specified; the finite tests retain their stated regulator and model domains.